

Introduction to Analysis of Binary Outcomes

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Eric S. Kawaguchi

Division of Biostatistics and Health Data Science
Department of Population and Public Health Sciences
University of Southern California

Binary outcomes

- Define a binary outcome by coding

$$Y = \begin{cases} 1 & \text{if event occurs} \\ 0 & \text{otherwise.} \end{cases}$$

- The outcome might inherently be dichotomous.
 - Example: disease vs. no disease, response vs. non-response
- A dichotomous outcome could be created by applying a threshold to a continuous measurement.
 - Example: Hypertension

$$Y = \begin{cases} 1, & \text{SBP} \geq 140\text{mmHg} \\ 0, & \text{SBP} < 140\text{mmHg} \end{cases}$$

Measures of frequency

- When Y is binary, our primary quantity of interest is the proportion (or probability) of the event:

$$p = \Pr(Y = 1)$$

- If Y is coded 0/1, then the sample proportion, $\hat{p}$, can be estimated as:

$$\hat{p} = \frac{\sum_{i=1}^n Y_i}{n} = \bar{Y}$$

- The sample proportion is just the sample mean!
- The sample proportion is an estimate of the population proportion.

Measures of frequency

- In many epidemiological studies, we often follow disease-free individuals for a fixed period of time.
- When Y is coding the presence or absence of a disease within that interval, the probability of developing the disease during the follow-up period is called the disease **risk***.
- The observed proportion of individuals who develop the disease is an estimate of the disease risk.
- Formally, risk is defined over a specified follow-up period.

*I'll be using risk instead of proportion moving forward.

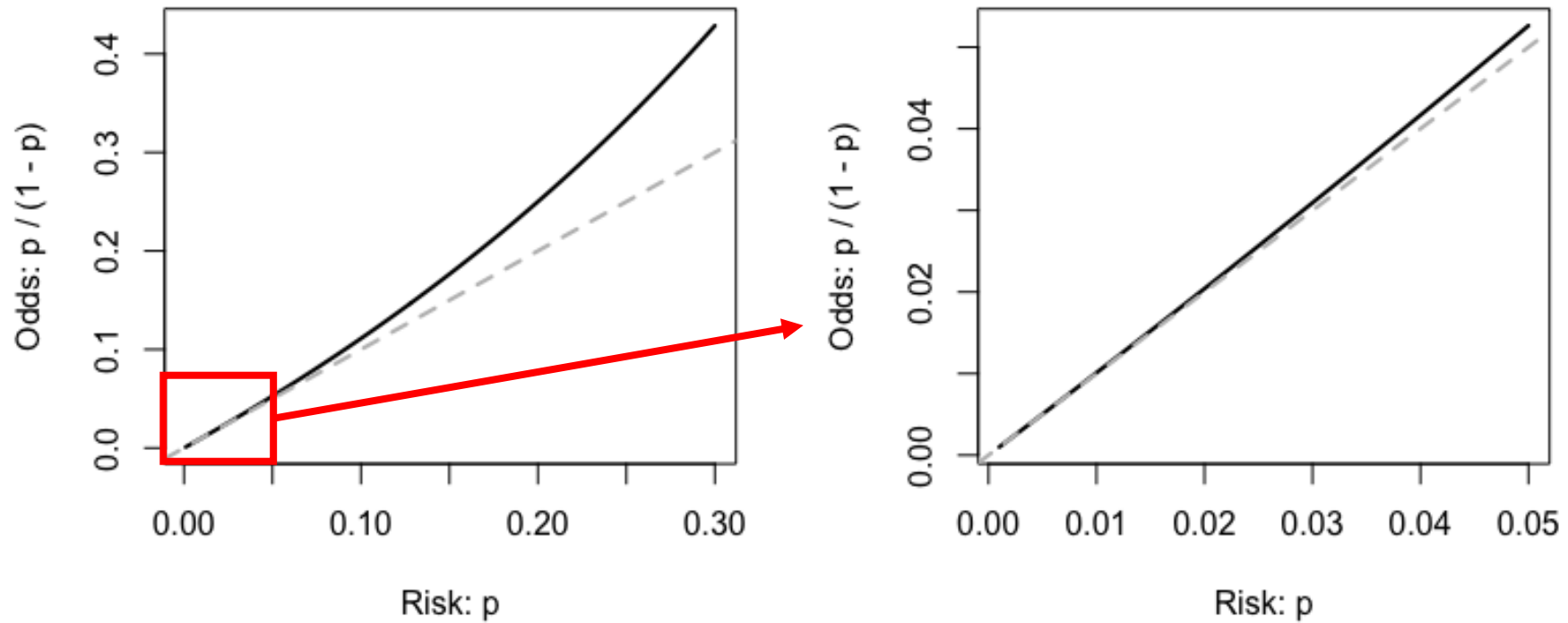
Measures of frequency

- A related quantity to risk is the odds, which compares the probability of the event to the probability of no event

$$\omega = \frac{\Pr(Y = 1)}{\Pr(Y = 0)} = \frac{p}{1 - p}$$

- If disease risk $p = 0.2$, then the odds are $\omega = 0.2/0.8 = 0.25$.
- $\omega \in [0, \infty)$
- For rare events, $p \approx \omega$ since $1 - p \approx 1$
 - Suggests that for rare diseases, the odds of disease can *approximate* the disease risk.

Odds vs. Risk



Measures of association

- We often want to compare risk between groups.
- Suppose we have two groups

$$X = \begin{cases} 1 & \text{if exposed} \\ 0 & \text{if unexposed} \end{cases}$$

- Example: Smokers vs. Non-smokers, New intervention vs. standard care.
- Define
 - $p_1 = \Pr(Y = 1|X = 1)$: Risk of disease ($Y = 1$) in the exposed group ($X = 1$).
 - $p_0 = \Pr(Y = 1|X = 0)$: Risk of disease in the unexposed group ($X = 0$).
- Note p_0 and p_1 are *conditional* quantities.
- Conditional odds can be defined similarly.

Measures of association

Two common measures of association between Y and X are

1. Relative risk (risk ratio):

$$RR = \frac{p_1}{p_0} = \frac{\Pr(Y = 1|X = 1)}{\Pr(Y = 1|X = 0)}$$

2. Odds ratio:

$$OR = \frac{\frac{p_1}{1 - p_1}}{\frac{p_0}{1 - p_0}}$$

- **RR = 2:** the risk of disease is twice as high in the exposed group as in the unexposed group.
- **OR = 2:** the odds of disease are twice as high in the exposed group as in the unexposed group.

Example

Given a 2x2 table:

	$X = 1$	$X = 0$	
$Y = 1$	a	b	$a + b$
$Y = 0$	c	d	$c + d$
	$a + c$	$b + d$	n

- $\hat{p}_0 = b/(b + d)$
- $\hat{p}_1 = a/(a + c)$
- Relative risk:

$$RR = \frac{\hat{p}_1}{\hat{p}_0} = \frac{\frac{a}{a + c}}{\frac{b}{b + d}}$$

- Odds ratio:

$$OR = \frac{\frac{\hat{p}_1}{1 - \hat{p}_1}}{\frac{\hat{p}_0}{1 - \hat{p}_0}} = \frac{ad}{bc}$$

Example (CHS)

Children's Health Study:

- Primary outcome: Incident asthma case (Yes/No)
- Exposure: Biological sex (Female/Male)

	Male	Female	
Yes	110	99	209
No	369	418	787
	517	479	

- Question: What is the association between asthma and biological sex?

Example (CHS)

- Estimated risk ratio (males relative to females):

$$\widehat{RR} = \frac{110/517}{99/479} \approx 1.03$$

- Interpretation: The risk of having asthma is 3% higher in males relative to females.

Example (CHS)

- Estimated odds ratio (males relative to females):

$$\widehat{OR} = \frac{110 \times 418}{369 \times 99} \approx 1.26$$

- Interpretation: The odds of having asthma is 26% higher in males relative to females.
- Notice that the odds ratio (1.26) is larger than the relative risk (1.03).
- We have that:

$$OR = RR \times \frac{1 - p_0}{1 - p_1} = RR \times \frac{\Pr(D = 0|E = 0)}{\Pr(D = 0|E = 1)}$$

RR vs. OR

- For a given 2×2 table:
 - $RR = 1$ if and only if $OR = 1$ (no association between X and Y)
 - $OR < RR < 1$
 - $OR > RR > 1$
- Direction of effect is consistent, but we can't say anything about the magnitude.
- Thus, if we use the OR to approximate the RR, then our estimated effect size will be optimistically biased (biased away from the null).
- The OR can approximate the RR if we have a *rare* disease/event

Example

	Male	Female	
Yes	110	99	209
No	369	418	787
	517	479	996



	Male	Female	
Yes	110	99	209
No	3690	4180	7870
	3800	4279	8079

Given a 2x2 table:

- $\widehat{RR} = \frac{110/517}{99/479} \approx 1.03$
- $\widehat{OR} = \frac{110 \times 418}{369 \times 99} \approx 1.26$

Given a 2x2 table:

- $\widehat{RR} = \frac{110/3800}{99/4279} \approx 1.25$
- $\widehat{OR} = \frac{110 \times 4180}{3690 \times 99} \approx 1.26$

RR vs. OR

Feng et. al. (2019) illustrated that a larger OR does not imply a larger RR:

Exposure 1: RR = 3; OR = 5

	$E = 1$	$E = 0$
$D = 1$	60	20
$D = 0$	60	100
	120	120

Exposure 2: RR = 2; OR = 7

	$E = 1$	$E = 0$
$D = 1$	100	50
$D = 0$	20	70
	120	120

Odds ratio

- In general, the RR (risk ratio) is easier to interpret than the OR.
- Why even deal with odds and odds ratios?
- The ability to estimate risk (and the risk ratio) depends on the sampling design.

Aside: Case-control study

- Suppose we want to identify risk factors for a rare disease (e.g., pancreatic cancer)
- Following a disease-free cohort long enough to get a sufficient number of events to study can be:
 1. Financially costly
 2. Time consuming
 3. Inefficient

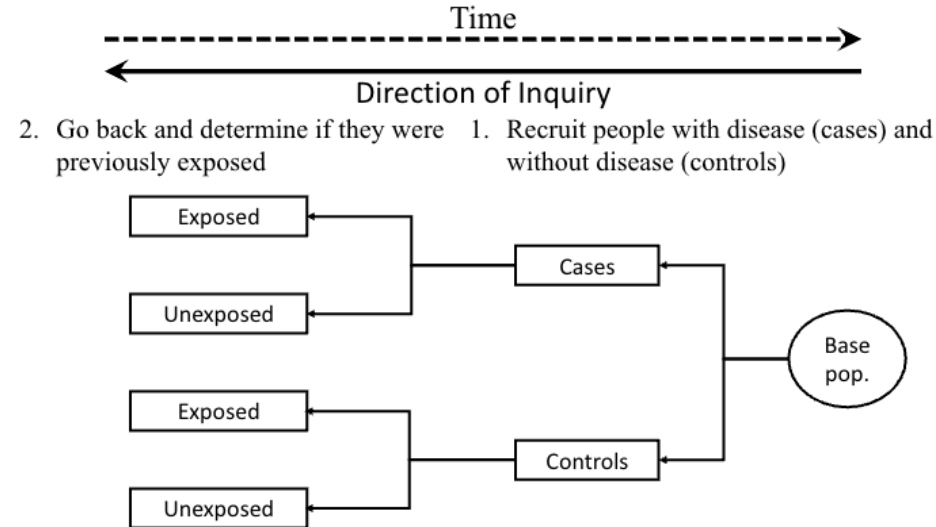
Example: Suppose a disease affects 5 in 10,000 people.

- To observe 100 cases, we might need to follow over 200,000 individuals.

Aside: Case-control study

Case-control studies provide an alternative approach:

1. Recruit individuals who already have the disease (cases)
2. Recruit a group of individuals who are disease free (controls)
3. Ascertain exposure measurements



Aside: Case-control study

- While case-control studies are efficient and cost effective for studying rare diseases, we over sampled the amount of people with the disease.
- We cannot estimate the risk, odds, and risk ratio from the observed data because our sample of those with the disease is not representative of our population.
- However, surprisingly, the odds ratio is still valid!

Example (CHS)

	Male	Female	
Yes	110	99	209
No	369	418	787
	517	479	996

- Estimated $\widehat{OR} = \frac{110 \times 418}{369 \times 99} \approx 1.26$
- Suggests that males have a higher odds of having asthma.
- Is the observed association greater than expected by chance alone?
- Test $H_0: OR = 1$ vs. $H_1: OR \neq 1$
 - This is also a test of $H_0: RR = 1$ vs. $H_1: RR \neq 1$ since $OR = 1$ if and only if $RR = 1$.

Pearson's χ^2 test

A common test for this hypothesis is the Pearson's χ^2 test.

Observed

	$X = 1$	$X = 0$	
$Y = 1$	O_{11}	O_{10}	$n_{1.}$
$Y = 0$	O_{01}	O_{00}	$n_{0.}$
	$n_{.1}$	$n_{.0}$	$n_{..}$

Expected

	$X = 1$	$X = 0$	
$Y = 1$	E_{11}	E_{10}	$n_{1.}$
$Y = 0$	E_{01}	E_{00}	$n_{0.}$
	$n_{.1}$	$n_{.0}$	$n_{..}$

- For $E = i, D = j$, O_{ij} is the observed counts and $E_{ij} = n_{i.}n_{.j}/n_{..}$ is the expected value *under* H_0 (no association).
- Basically, answering the question: *Are the observed counts sufficiently different from the expected counts to reject the null hypothesis?*

Pearson's χ^2 test

- The Pearson's χ^2 test statistic is defined as:

$$X^2 = \sum_{ij} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \stackrel{H_0}{\sim} \chi^2(1),$$

where we take the sum over all the cells (4 for a 2×2 table).

- Large discrepancies between observed and expected counts produce large values of X^2 , providing evidence *against* the null hypothesis.
- The χ^2 test is a *two-sided test* even though critical region is one-sided.
- This procedure is based on the normal approximation (i.e., central limit theorem).
- Rules of thumb: No expected cell is less than 5 and no observed cell is zero.

Example (CHS)

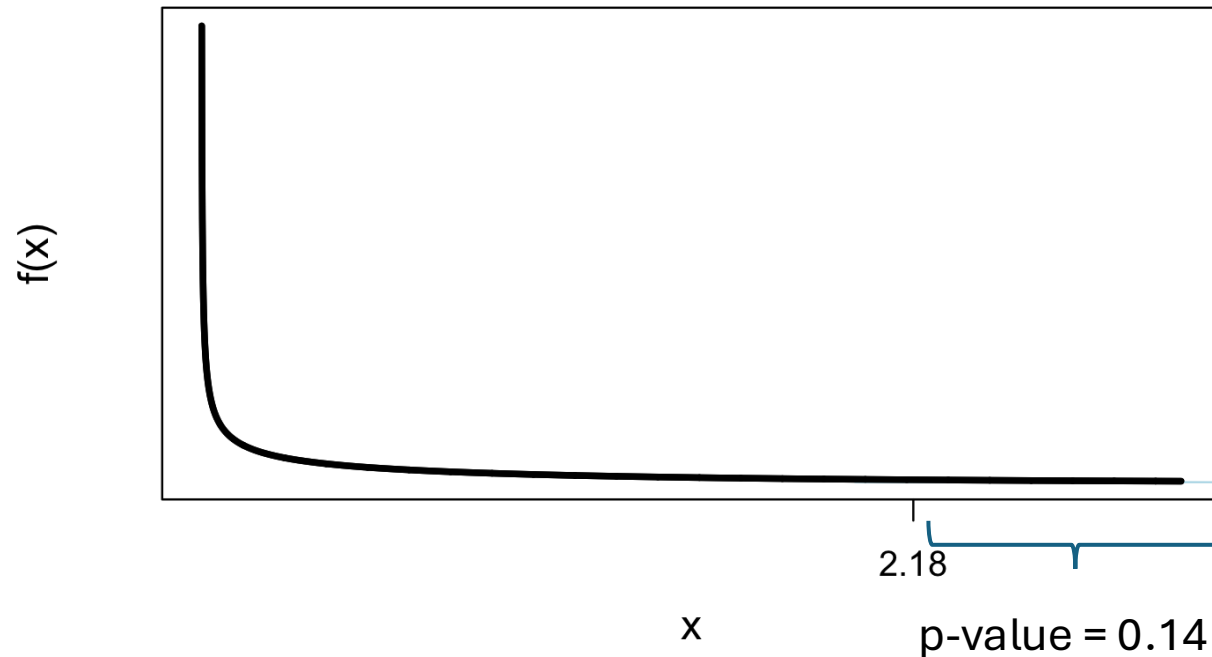
Observed			
	Male	Female	
Yes	110	99	209
No	369	418	787
	517	479	996

Expected			
	Male	Female	
Yes	108.49	100.51	209
No	408.51	378.49	787
	517	479	996

- $E_{11} = \frac{517 \times 209}{996} = 108.49$
- $E_{10} = \frac{479 \times 209}{996} = 100.51$
- Note $E_{11} + E_{10} = 108.49 + 100.51 = 209$

Example (CHS)

$$\chi^2 = \sum_{ij} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \frac{(110 - 108.49)^2}{108.49} + \dots + \frac{(418 - 378.49)^2}{378.49} = 2.18$$



Pearson's χ^2 test

```
> chisq.test(matrix(c(110, 369, 99, 418), nrow = 2, ncol =  
2), correct = FALSE)
```

Pearson's Chi-squared test

```
data:  matrix(c(110, 369, 99, 418), nrow = 2, ncol = 2)
```

```
X-squared = 2.1832, df = 1, p-value = 0.1395
```

We do not have sufficient evidence to conclude that biological sex is associated with asthma ($p = 0.14$).

Pearson's χ^2 test

- Pearson's χ^2 test can also be used when the exposure has $K > 2$ categories.
 - Example: BMI (Underweight, Healthy, Overweight, Obese)

	BMI				
	Underweight	Healthy	Overweight	Obese	
Yes					
No					

- Same idea as before: Compared observed and expected counts in every cell.
- Tests for an overall association between the exposure and outcome.
 - Rejecting H_0 tells us that at least one BMI group differs, but not which group(s) differ.

Logistic Regression

Example (CHS)

- What if we want to see if child's age was associated with risk of asthma?
- If child's age is continuously measured, data cannot be laid out in a 2×2 table.
- We can dichotomize age, say < 10 ($X = 0$) and ≥ 10 ($X = 1$), but we'll be losing some valuable information.

Logistic regression

- Linear regression: We model the *mean* of our continuous outcome Y .
- For a binary outcome Y , coded $\{0, 1\}$, the *mean* of Y is simply the probability of “event”, $\Pr(Y = 1)$.
- In parallel to linear regression, we can model the probability of “event” given the value of $X = x$,

$$\Pr(Y = 1|X = x) = p = \alpha + \beta x$$

Logistic regression

- Issue 1: p is supposed to be a *probability*, so it needs to be between 0 and 1.
 - We place no restriction on α and β .
 - $\alpha + \beta x$ can take on any value between $-\infty$ and ∞ (especially if x is continuous)
 - Question: Can we “match” the right-hand side, $\alpha + \beta x$, to some function of the left-hand side?
- Issue 2: Sometimes we can't reliably estimate risk!
 - Maybe we can model odds ratios?

Logistic regression

- Odds: For $p \in (0, 1)$

$$\omega = \frac{p}{1 - p}$$

- Logit: For $p \in (0, 1)$

$$\text{logit}(p) = \log\left(\frac{p}{1 - p}\right) = \log(\omega)$$

- The logit is also referred to as the “log odds”.
- The odds maps values from $(0, 1)$ to $(0, \infty)$, whereas the logit maps values from $(0, 1)$ to $(-\infty, \infty)$.

Logistic regression

- The (simple) logistic regression model can be written as follows:

$$\text{logit}(\pi(x)) = \log\left(\frac{\Pr(Y = 1|X = x)}{1 - \Pr(Y = 1|X = x)}\right) = \beta_0 + \beta_1 x.$$

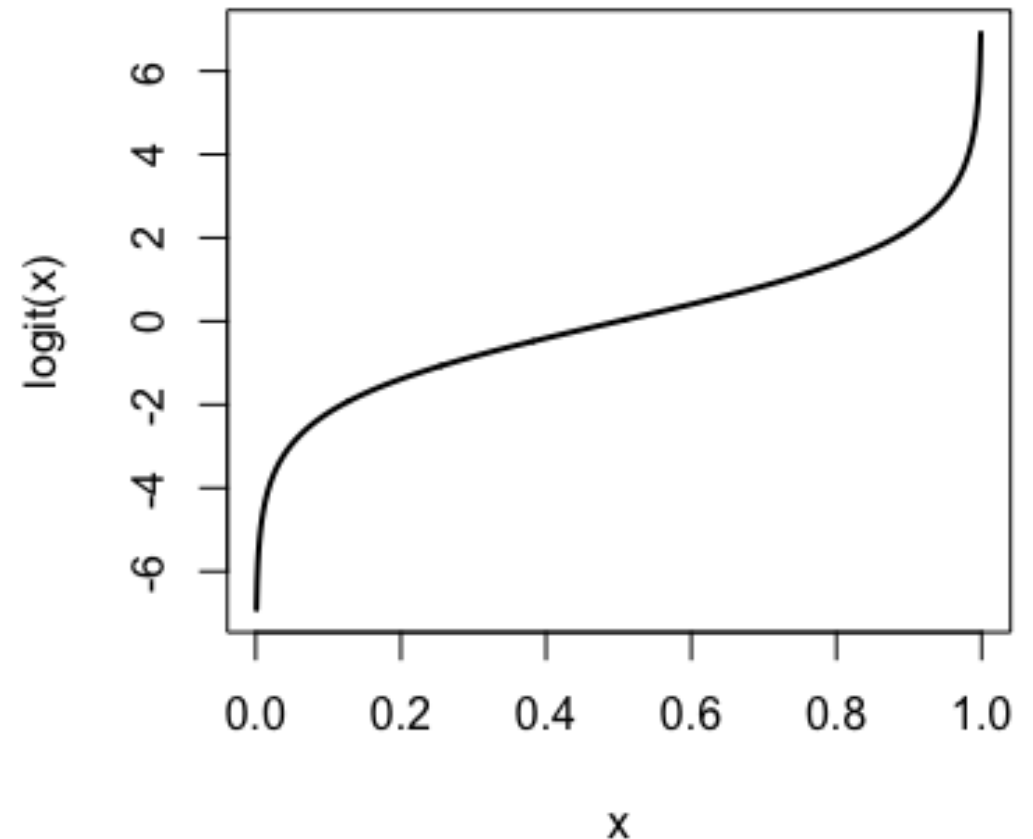
- $\pi(x)$ is the probability of disease (risk) given $X = x$.
- We assume linearity on the *logit* (i.e., log-odds) scale.
- It can be shown using algebra that

$$\Pr(Y = 1|X = x) = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)}$$

Logistic regression

- The inverse of the logit function is the *standard logistic function*:

$$\sigma(x) = \frac{1}{1 + \exp(-x)} = \frac{\exp(x)}{1 + \exp(x)}.$$



Interpreting the coefficients: β_0

- Plug in $X = 0$,

$$\text{logit}(\pi(0)) = \log\left(\frac{\Pr(Y = 1|X = 0)}{1 - \Pr(Y = 1|X = 0)}\right) = \beta_0 + \beta_1 \times 0 = \beta_0$$

- Exponentiating both sides yields:

$$\frac{\Pr(Y = 1|X = 0)}{1 - \Pr(Y = 1|X = 0)} = \exp(\beta_0).$$

- Thus, $\exp(\beta_0)$ is the odds of disease ($Y = 1$) when $X = 0$.

Interpreting the coefficients: β_1

- Plug in $X = a$,

$$\text{logit}(\pi(a)) = \log\left(\frac{\Pr(Y = 1|X = a)}{1 - \Pr(Y = 1|X = a)}\right) = \beta_0 + \beta_1 a$$

- Plug in $X = b$,

$$\text{logit}(\pi(b)) = \log\left(\frac{\Pr(Y = 1|X = b)}{1 - \Pr(Y = 1|X = b)}\right) = \beta_0 + \beta_1 b$$

- The difference yields

$$\text{logit}(\pi(b)) - \text{logit}(\pi(a)) = \beta_0 + \beta_1 b - [\beta_0 + \beta_1 a] = \beta_1(b - a)$$

Interpreting the coefficients: β_1

- One can show that

$$\text{logit}(\pi(b)) - \text{logit}(\pi(a)) = \log \left(\frac{\frac{\Pr(Y = 1|X = b)}{1 - \Pr(Y = 1|X = b)}}{\frac{\Pr(Y = 1|X = a)}{1 - \Pr(Y = 1|X = a)}} \right).$$

- The *difference of the log odds (logit)* is just the *log of the odds ratio*.
- Therefore, $\exp(\beta_1(b - a))$ is the odds ratio for a subject exposed at level b , relative to a persons exposed at level a .
 - Often, we set $b = a + 1$, so that $\exp(\beta_1(a + 1 - a)) = \exp(\beta_1)$ is the OR per unit increase of the exposure.

Some notes:

For binary exposures:

- When X is binary, $\exp(\beta_1)$ is the OR between the exposed ($X = 1$) and unexposed ($X = 0$) groups.

$$\frac{\frac{\Pr(Y = 1|X = 1)}{1 - \Pr(Y = 1|X = 1)}}{\frac{\Pr(Y = 1|X = 0)}{1 - \Pr(Y = 1|X = 0)}} = \exp(\beta_1) \Rightarrow \frac{\Pr(Y = 1|X = 1)}{1 - \Pr(Y = 1|X = 1)} = \exp(\beta_1) \frac{\Pr(Y = 1|X = 0)}{1 - \Pr(Y = 1|X = 0)}$$


Odds of disease when $X = 1$

Odds of disease when $X = 0$

Hypothesis testing

- $\beta_1 / \exp(\beta_1)$ quantifies the association between X and Y .
- We want to test if this association is statistically significant.
 - $H_0: \beta_1 = 0$ vs. $H_1: \beta_1 \neq 0$ or equivalently $H_0: OR = 1$ vs. $H_1: OR \neq 1$
- One way to test this hypothesis is to compare the test statistic:

$$Z = \frac{|\hat{\beta}_1|}{\widehat{SE}(\hat{\beta}_1)} \stackrel{H_0}{\sim} N(0, 1)$$

- We reject H_0 if Z is large, or equivalently, if the p-value is small compared to some prespecified significant threshold.

Logistic regression in R

- Logistic regression can be fit in R using the `glm` function.
- `glm`: **Generalized linear model**

```
glm(y ~ x, family = "binomial", data = d)
```

Logistic regression in R

```
lr_sex <- glm(asthma ~ sex, family="binomial", data=chs)
summary(lr_sex)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.4404	0.1118	-12.886	<2e-16 ***
sexM	0.2300	0.1559	1.476	0.14

- $\hat{\beta}_1 = 0.23$, $\exp(\hat{\beta}_1) = 1.258$.
 - p-value = 0.14, not statistically significant at the $\alpha = 0.05$ level.
- $\hat{\beta}_0 = -1.44$, $\exp(\hat{\beta}_0) = 0.236$.

Example (CHS)

	Male	Female	
Yes	110	99	209
No	369	418	787
	517	479	

- $OR = \frac{418 \times 110}{99 \times 369} = 1.258$
- Odds of asthma for females: $\hat{\pi}_0 = \frac{\frac{99}{517}}{\frac{418}{517}} = \frac{99}{418} = 0.236$
- Same results as the logistic regression model!

Logistic regression in R

```
lr_age <- glm(asthma ~ age, family="binomial", data=chs)
summary(lr_age)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.7521	1.9465	0.386	0.699
age	-0.1364	0.1278	-1.067	0.286

- $\exp(\hat{\beta}_1) = \exp(-0.136) = 0.87$.
 - A one-year increase in age is associated with a 13% decrease in the odds of having asthma.
- $\exp(\hat{\beta}_0) = \exp(0.7521) = 2.12$

Logistic regression in R

For regression with several predictors (multivariable logistic regression):

$$\text{logit}(\pi(x_1, x_2, x_3)) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3.$$

- $\exp(\beta_1)$ is the odds ratio for Y associated with a one-unit change in x_1 while keeping all other variables (x_2 and x_3) fixed.
- $\exp(\beta_2)$ is the odds ratio for Y associated with a one-unit change in x_2 while keeping all other variables (x_1 and x_3) fixed.
- $\exp(\beta_3)$ is the odds ratio for Y associated with a one-unit change in x_3 while keeping all other variables (x_1 and x_2) fixed.

Logistic regression in R

```
lr_sex_age = glm(asthma ~ sex + age, family = "binomial", data = chs)
summary(lr_sex_age)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.9373	1.9517	0.480	0.631
sexM	0.2491	0.1568	1.589	0.112
age	-0.1567	0.1285	-1.219	0.223

- $\exp(\hat{\beta}_{age}) = \exp(-0.1567) = 0.85$
 - One-year increase in age is associated with a 15% decrease in the odds of having asthma after adjusting for child's biological sex.

Some notes

- Logistic regression is the most commonly-used regression model for binary outcomes.
- Coefficient estimates are interpreted as log odds-ratios.
- Odds ratios can be estimated in all study designs.
(case-control, cross sectional, cohort, etc.)
- When the disease is rare in the general population, the odds ratio approximates the relative risk ratio.

Extension to Multinomial Logistic Regression

When the outcome has more than two *unordered* categories.

Example: Cancer type (Breast, Lung, CRC)

Model idea: Model the log-odds of each category versus a reference category.

$$\log\left(\frac{\Pr(Y = k|X)}{\Pr(Y = \text{Reference}|X)}\right) = \beta_{0k} + \boldsymbol{\beta}_k^T X, \quad k = 1, \dots, K - 1$$

$\exp(\beta_{kj})$ is the odds ratio for being in category k , relative to the reference category, associated with a one-unit increase in X_j .

Extension to Ordinal Logistic Regression

When the outcome has more than two *ordered* categories.

Example: Disease severity

Model idea (proportional odds model): Model cumulative probabilities

$$\log\left(\frac{\Pr(Y > k|X)}{\Pr(Y \leq k|X)}\right) = \beta_{0k} + \boldsymbol{\beta}^T X, \quad k = 1, \dots, K - 1$$

$\exp(\beta_j)$ is the *common* odds ratio for being in a higher category associated with a one-unit increase in X_j .